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Non-surgical periodontal therapy affects metabolic control in diabetics: a randomized controlled clinical trial

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ABSTRACT

Background: Periodontal diseases and diabetes are two common diseases with high prevalence. Many clinicians have accepted the relationship between these two diseases. Some investigators have reported that periodontal treatment may enhance the metabolic control of diabetes. The effects of non-surgical periodontal treatment on metabolic control in people with type 2 diabetes mellitus (DM2) were examined.
Methods: Forty patients with DM2 and chronic periodontitis [mean age = 50.29 years; mean glycated haemoglobin (HbA1c) = 8.72] were randomly assigned to two groups. The treatment group (n = 22) received full-mouth scaling and root planing, whereas the control group (n = 18) received no periodontal treatment. Gingival index (GI), plaque index (PI), probing pocket depth (PPD), clinical attachment level (CAL), fasting plasma glucose (FPG), HbA1c, total cholesterol (TC), triglyceride (TG) and cholesterol levels were recorded at baseline and compared to data collected three months later.
Results: The groups did not differ in gender ratio, age or clinical parameters [PPD (p = 0.107), CAL (p = 0.888), PI (p = 0.180)] and biochemical markers at baseline [FPG (p = 0.429), HbA1c (p = 0.304), TG (p = 0.486), TC (p = 0.942), LDL (p = 0.856) and HDL (p = 0.881)]. FPG, HbA1c and clinical parameters differed between the treatment and control groups (p = 0.006, 0.003 and 0, respectively). From baseline to follow-up (after three months), HbA1c levels decreased in the treated group (p = 0.003). In the same time period, FPG, GI, PPD and CAL increased in the control group (p = 0.016, 0.0, 0.0 and 0.004, respectively) but HbA1c did not change significantly.
Conclusions: Non-surgical periodontal therapy could improve metabolic control in diabetic patients.

Keywords: Diabetes mellitus type 2, periodontal disease, HbA1c protein, non-surgical periodontal debridement, dental scaling, root planing.

Abbreviations and acronyms: CAL = clinical attachment loss; DM2 = diabetes mellitus type 2; FPG = fasting plasma glucose; HDL = high density lipoprotein cholesterol; LDL = low density lipoprotein cholesterol; PI = plaque index; PPD = probing pocket depth; TC = total cholesterol; TG = triglycerides; TNF = tumour necrosis factor.

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INTRODUCTION

Type 2 diabetes mellitus (DM2), the most common type of diabetes, is characterized by hyperglycaemia, hyperlipidaemia and associated complications. The classic major complications of diabetes are microangiopathy, nephropathy, neuropathy, retinopathy, macrovascular disease, delayed wound healing and periodontitis.[1] The hyperlipidaemia that accompanies hyperglycaemia involves marked elevation of low density lipoprotein cholesterol (LDL), triglycerides (TG) and omega-6 free fatty acids.[2,3] This abnormality in fatty acid metabolism is thought to underlie the development of many diabetic complications.[4]

One of the leading complications of diabetes, periodontitis, is an infection of the periodontal support tissues.[5] There is an interacting, complex relationship between diabetes and periodontitis. Many studies have shown a greater incidence and a greater severity of periodontitis in diabetic patients.[6–9] Meanwhile, a number of studies have suggested that periodontitis may actually be a risk factor for diabetic complications as well.[1,10–15] Southerland *et al.*[16] proposed a common pathogenesis involving an increased inflammatory response for periodontitis and diabetes. Indeed, patients with periodontitis have increased serum levels of inflammatory cytokines, while diabetic patients have hyperinflammatory immune cells that can aggravate

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the increased production of inflammatory cytokines.[17] This exacerbation can increase insulin resistance and make it more difficult for patients to control their diabetes.[17,18]

Over a decade ago Christgau *et al.*[19] demonstrated that patients' keratin, TG and total cholesterol (TC) can be affected to a small degree by periodontal therapy. Observations that periodontal therapy appears to reduce periodontal infection and inflammation suggest that periodontal therapy may facilitate metabolic control of diabetes, improving insulin sensitivity by reducing peripheral inflammatory cytokine levels.[17,20] Indeed, among the earlier studies that investigated the effects of periodontal therapy on glycaemic control, a large number reported improved glycaemic control after periodontal treatment,[2,21–27] while others did not find a positive effect of periodontal therapy on glycaemic control.[11,16,18,28–30] Stewart *et al.*[21] previously reported a decrease in HbA1c level following non-surgical periodontal treatment in diabetic patients, but that study did not compare periodontal parameters to control group data and included patients whose medication regimen changed during the study period. The present study was performed to investigate the effects of non-surgical periodontal therapy on metabolic control in DM2 patients. It was a randomized, controlled clinical trial and included a group of DM2 control subjects who did not receive periodontal treatment during the study period.

MATERIALS AND METHODS

Design

This study was a randomized, controlled clinical trial conducted between June 2007 and September 2008 in Mashhad, Iran.

Patients

Based on studies by Rodrigues *et al.*[2] and Kiran *et al.*[37] (with $\alpha = 0.05$ and $\beta = 0.2$), sample size was calculated as 20 patients in each group. The protocol employed for this study was approved by the Research Council and the Ethical Committee of Mashhad University of Medical Sciences, Iran. This trial was registered at http://www.clinicaltrial.gov and the clinicaltrial.gov identifier was NCT01252082. Eligible patients were selected and recruited from the Parsian Diabetics Clinic and the Mashhad Diabetics Center (Mashhad, Iran). They were then referred to the Periodontics Department of Mashhad Dental School in Iran. Fifty-six patients with DM2 and periodontitis were assessed, and a screening oral examination and medical history review carried out to confirm eligibility. Sixteen patients chose to leave the study; eight did not meet the inclusion criteria and the other eight refused to participate. Finally, 40 subjects gave written informed consent and completed the study. DM2 subjects were selected for this study because DM2 is highly prevalent in the general population and its incidence is increasing.[31]

The inclusion criteria were as follows: (1) mild to moderate periodontitis according to the criteria of the American Academy of Periodontology;[32] (2) diagnosis of DM2 with glycated haemoglobin (HbA1c) values over 7%; (3) no major diabetic complications; (4) blood sugar controlled with glyben glamide and metformin, without insulin administration; and (5) no systemic antibiotic administration or periodontal treatment within the last six months.

The following exclusion criteria were applied: (1) presence of systemic diseases other than DM2 that could influence the course of periodontal disease; (2) intake of immunosuppressive drugs, steroids, hydantoin or non-steroidal anti-inflammatory drugs; (3) tobacco use; (4) pregnancy or intention to become pregnant during the study period; (5) fixed orthodontic appliances; and (6) refusal or inability to give informed consent.

Patients were randomly divided into treatment and non-treatment (control) groups by the study research assistant (KK) using a computer generated random numbers table. All the patients were under strict medical supervision and, as a prerequisite, no additional guidance in managing diabetic status or changes in diet, medication or physical therapy was given so that we could directly evaluate the effects of periodontal treatment on metabolic control.

Peridontal assessments

All subjects underwent a comprehensive clinical examination by a single examiner (AMT) who was an expert periodontist and blinded to the subjects' group assignment. Intra-examiner reliability was tested by examining three patients in an identical manner one hour apart and observing more than 95% of recordings being within 1 mm. Periodontal parameters were recorded at baseline (day 0) and three months later in both groups. Severity of periodontal disease was evaluated using gingival index (GI), plaque index (PI), clinical attachment level (CAL) and probing pocket depth (PPD). PI was quantified using the O'Leary index in which the four smooth surfaces of all existing teeth were scored and a percentage of surfaces with plaque was recorded.[33] Gingival status recordings were made for each tooth according to established GI criteria.[34] PPD was measured from the gingival margin with a Williams' periodontal probe (Hu-Friedy, Chicago, IL, USA). CAL was defined as the distance

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from the cemento-enamel junction to the bottom of the pocket.

Metabolic assessments

Venous blood samples were taken from each patient and analysed for fasting plasma glucose (FPG), HbA1c, TG, TC, high density lipoprotein cholesterol (HDL) and LDL. Metabolic measurements were performed at baseline and three months later in both groups. All laboratory assessments were performed in the Pars Laboratory (Mashhad, Iran) using Cobas Integra 700 apparatus (Roche Diagnostics, Germany). Pars Laboratory is a well-known private laboratory where all diabetic patients are referred to because of its high standards. Reliability of biochemical measurements were confirmed through credentialed regulations and shown to be within acceptable standards.

Periodontal treatment

All periodontal treatments were performed by a general practitioner (YB) under the supervision of an expert periodontist (HRA). Prior to the first treatment session, patients in both groups received standard oral hygiene instructions, placement of emergency restorations and extraction of unsalvageable teeth. In the treatment group, patients were given local anaesthetic (Lidocaine HCl 2% and Epinephrine 1:100 000 injection, 3–5 single dose cartridges of 1.7 mL; Daroopakhsh, Iran) and then subjected to full-mouth scaling and root planing by using an ultrasonic device (VGE 302k, Juya Electric Co., Tehran, Iran) and standard periodontal curettes with no time limitation. Patients did not receive any further periodontal treatment for three months and medical treatment for diabetes remained unchanged. Control group patients received no periodontal treatment during the study period but after completion of the study, for ethical reasons, they received a full non-surgical and supportive periodontal treatment. Three months after the baseline examinations, all 40 subjects were recalled for a second clinical examination and all parameters that were assessed at baseline were reassessed.

Statistical analysis

Statistical analyses were performed using SPSS 11.5. In each of the analyses, $p < 0.05$ was regarded as significant. There were no dropouts in both groups so we used per protocol analysis. Data from the baseline and three-month follow-up evaluations were subjected to the Kolmogorov-Smirnov test to determine whether they were normally distributed. Means and standard deviations were calculated for each variable at each time point for each group. Significant group differences over time were calculated using Student's t-test analysis for continuous data (PPD, CAL, FBS, TG, TC, LDL, HDL) and the Mann-Whitney test for non-parametric data (PI, GI and HbA1c). Differences between baseline and three-month follow-up data for each group were calculated using paired t-tests for continuous data. Wilcoxon's rank sum test was used to compare non-parametric data values between baseline and the three-month follow-up time point. To make an association measurement between periodontal therapy and metabolic control, we calculated number of well-controlled unchanged and poorly controlled patients in both groups after three months. We considered patients with HbA1c <7% as well-controlled and HbA1c ≥7% as poorly controlled.[17] Chi-square test was used to show this association and logistic regression test was performed to evaluate the effect of baseline periodontal parameter (PD) and periodontal therapy on metabolic control.

RESULTS

The subject cohort comprised of 20 (50%) females and 20 (50%) males, with an overall mean age of 50.29 ± 3 years. The mean age of females (48.1 ± 3 years) was similar to that of males (52.48 ± 3 years; $p = 0.9$). The distribution of patients with respect to gender was similar between the groups, with the treatment group consisting of 13 (59.1%) females and 9 (40.9%) males and the control group consisting of 7 (38.9%) females and 11 (61.1%) males ($p = 0.341$). In 2 patients we had to extract hopeless teeth; 1 in the treated group with an unrestorable third molar and 1 in the control group with a periodontally hopeless premolar. One patient in the treatment group suffering from an endodontic problem received required treatment.

Baseline assessments

All assessed variables were similar between control and treated groups at baseline: PPD ($p = 0.107$), CAL ($p = 0.888$), PI ($p = 0.180$), FPG ($p = 0.429$), HbA1c ($p = 0.304$), TG ($p = 0.486$), TC ($p = 0.942$), LDL ($p = 0.856$) and HDL ($p = 0.881$).

Effects of treatment on periodontal assessments

Periodontal therapy significantly improved all periodontal parameters in the treated group (Table 1). For the treated group, from the baseline assessment to the three-month follow-up assessment, PPD decreased from 2.31 ± 1.3 to 2.21 ± 1.2 ($p = 0.012$), CAL decreased from 3.14 to 2.8 ($p = 0.00$), PI decreased from 88.9% to 63.22% ($p = 0.005$) and GI decreased from 1.86 ± 0.83 to 1.24 ± 1.03 ($p = 0.00$). In the same time period, PPD ($p = 0.00$), GI ($p = 0.04$) and CAL

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Table 1. Comparison of periodontal parameters (mean ± SD and mean change) in treated and control groups

Group	n	Periodontal parameter	Pre-tx	Post-tx	P vs. pre-tx	Mean change	P vs. control
Control		PPD (mm)	2.06 ± 0.24	2.33 ± 0.3	***0.00***	0.27	
	18	CAL (mm)	3.1 ± 1.05	3.47 ± 1.44	***0.004***	0.37	
		PI (%)	94.44 ± 6.62	87 ± 18.7	***0.06***	−7.44	
		GI	1.15 ± 0.51	1.723 ± 0.48	***0.049***	0.57	
Treatment		PPD (mm)	2.31 ± 0.65	2.21 ± 0.6	***0.012***	−0.1	***0.00***
	22	CAL (mm)	3.14 ± 1.08	2.8 ± 1.09	***0.00***	−0.34	***0.00***
		PI (%)	88.9 ± 17.38	63.22 ± 21.13	***0.005***	−25.68	***0.002***
		GI	1.867 ± 0.83	1.24 ± 1.03	***0.00***	−0.62	***0.00***

tx = treatment; P = p-value.

Table 2. Comparison of metabolic parameters (mean ± SD and mean change) in treated and control groups

Group	n	Metabolic parameter	Pre-tx	Post-tx	P vs. pre-tx	Mean change	P vs. control
Control		FPG (mg/dl)	159.22 ± 37.01	169 ± 38.95	***0.016***	9.78	
		HBA1c (%)	8.72 ± 2.22	8.97 ± 1.82	*0.263*	0.25	
	18	TG (mg/dl)	150.94 ± 27.21	147.44 ± 29.18	*0.706*	−3.5	
		TC (mg/dl)	192.44 ± 27.2	197.16 ± 27.1	*0.46*	4.72	
		LDL (mg/dl)	117.11 ± 34.17	114.55 ± 38.19	*0.776*	−2.56	
		HDL (mg/dl)	46.33 ± 12.39	43.88 ± 9.71	*0.55*	−2.45	
Treatment		FPG (mg/dl)	170.95 ± 52.4	153.45 ± 44.44	***0.006***	−17.5	***0.00***
		HBA1c (%)	8.15 ± 1.18	7.41 ± 1.18	***0.003***	−0.74	***0.003***
	22	TG (mg/dl)	137.81 ± 81.67	129.95 ± 57.88	*0.495*	−7.86	*0.773*
		TC (mg/dl)	191.81 ± 26.81	185.63 ± 30.92	*0.337*	−6.18	*0.231*
		LDL (mg/dl)	118.81 ± 24.9	112.68 ± 31.61	*0.309*	−6.13	*0.73*
		HDL (mg/dl)	45.81 ± 9.18	44.04 ± 6.7	*0.083*	−1.77	*0.861*

tx = treatment; P = p-value.

(p = 0.004) all increased in the control group, while PI remained similar between the two time points for controls (p = 0.06).

Effects of treatment on metabolic assessments

As shown in Table 2, patients in the treatment group exhibited a decrease in FPG from the baseline assessment (170.95 ± 52.4) to the three-month follow-up assessment (153.45 ± 44.44; p = 0.006). HbA1c levels decreased from 8.15 ± 1.18 to 7.41 ± 1.18 after treatment (p = 0.003). TG, TC, LDL and HDL were not altered by the treatment (p = 0.49, p = 0.33, p = 0.30 and p = 0.08, respectively). In the control group, FPG increased from 159.22 ± 37.01 to 169 ± 38.95 (p = 0.016), while HBA1c, TG, TC, HDL and LDL values remained not significantly changed after three months. Our data analysis showed that periodontal therapy was associated with decreased FPG and HbA1c levels, but no changes in plasma lipid levels (Table 2).

In the control group after three months, 5 patients were considered as well-controlled, 9 patients were poorly controlled and in 4 patients HbA1c remained unchanged (unchanged metabolic control). In the treatment group, 11 patients were well-controlled and 11 patients remained unchanged. Chi-square test showed a significant association (p = 0.001) between periodontal therapy and metabolic control (Table 3). Logistic regression analysis showed that regardless of pre-treatment pocket depth, periodontal therapy was associated with a decrease in HbA1c (p = 0.005).

Table 3. Association measurement between periodontal therapy and metabolic control

Group	N	Metabolic control		
		Well-controlled	Unchanged	Poorly controlled
Control	18	5	4	9
Treatment	22	11	11	

Well-controlled: HbA1c <7%.
Poorly controlled: HbA1c ≥7%.
Pearson Chi-square: 14.259, p = 0.001.

DISCUSSION

The present study showed significant decreases in HbA1c and FPG three months after non-surgical periodontal therapy in patients with DM2, while blood lipid parameters remained unchanged. These findings suggest that a reduction of periodontal infection can reduce HbA1c levels within a short period of time and thus may improve metabolic control in DM2 patients.

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The outcome of this study corroborates prior evidence supporting an interaction between periodontal status and diabetic metabolic control,[4,11,35] and supports the hypothesis that a successful periodontal treatment can improve glucose metabolism. These findings document a direct interrelationship between periodontal conditions and metabolic parameters in DM2 patients, extending earlier studies reporting a similar relationship that did not consider DM2 specifically.[28,36] Moreover, our findings are consistent with prior positive responses to non-surgical periodontal therapy in persons with DM2 reported by Westfelt *et al.*[28] and Kiran *et al.*[37] Our findings contradict those of Rodrigues *et al.*[2] who reported no difference in periodontal parameters in DM2 patients and healthy controls three months after periodontal therapy.

Periodontal disease may affect insulin signalling through pro-inflammatory mediators. The highly vascularized inflamed periodontium can be a source of inflammatory mediators, such as tumor necrosis factor (TNFα), which can affect glucose and fat metabolism.[38,39] The pro-inflammatory cytokine TNF impairs insulin signalling by increasing adipose secretion of free fatty acids. There is a consensus that this process strengthens glycaemic control in diabetic patients by raising insulin resistance. Accordingly, periodontal therapy might improve glycaemic control by decreasing pro-inflammatory mediators.

There is growing evidence in the literature supporting the notion that non-surgical periodontal therapy can improve glycaemic control. Kiran *et al.*[37] observed an improvement in periodontal parameters and a decrease in HbA1c levels three months after mechanical periodontal therapy, administered with no changes in medical therapy or diet, in DM2 patients with periodontitis. They suggested that the improvement in the HbA1c values could have been due to reduction in GI and bleeding on probing. Yang *et al.*[40] found decreases in TNF, HbA1c and periodontal parameters in DM2 patients after periodontal therapy, providing support for the possibility that periodontal therapy may reduce HbA1c values by reducing TNF concentrations in DM2 patients with periodontitis.

On the contrary, Promsudthi *et al.*[41] did not observe a reduction in HbA1c levels following mechanical periodontal treatment combined with systemic doxycycline. In contrast, Grossi *et al.*[10] and Iwamato *et al.*[22] reported decreases in HbA1c levels which they attributed to reduced TNF levels in circulation in response to systemic doxycycline, which decreased periodontal infection and inflammation.

In this study, we observed decreased HbA1c values in patients whose diabetes was moderate to poorly controlled. Ryan[42] suggested that more extensive periodontal treatment, such as surgery or local/systemic antibiotic therapy, may be needed in order to improve glycaemic control in poorly controlled diabetic patients. Indeed, many factors may influence the short-term glucose level of diabetic patients, including their medical care. To determine the relative contribution of periodontal therapy to glycaemic control, our study as a prerequisite included only patients who did not have any change in their diabetic control regimen during the three-month study period. We also found a significant association between periodontal therapy and metabolic control.

In the present study, all probing depths in both groups were measured at <5 mm. Therefore, unlike many other studies, highly variable periodontitis severity was not a confounding factor. Our analysis showed that the positive effect of periodontal therapy on metabolic control was irrespective of pre-treatment probing depth. Hence, the improved HbA1c measures seen after treatment can be attributed to diminished gingivitis.

There were two limitations in this study worth noting. Firstly, while the strict inclusion and exclusion criteria helped minimize confounding factors, it also limited the patient sample to a relatively small size. Secondly, we only examined subjects at a single three-month follow-up time point. Researchers have proposed different periods for reassessing periodontal parameters following non-surgical treatment. Morrison *et al.*[43] and Lowenguth *et al.*[44] suggested a one-month period while Badersten *et al.*[45] believed that maximum tissue reattachment in deepened periodontal pockets (>4 mm) occurs 4–5 months after treatment and can continue to increase for 12 months in 12 mm-deep periodontal pockets. Clarifying the effect of non-surgical periodontal therapy on glycaemic control will require further studies with larger sample sizes and more longitudinal data.

CONCLUSIONS

The present study demonstrated that non-surgical periodontal therapy can effectively decrease FPG and HbA1c levels in moderate to poorly controlled diabetic patients. Periodontal therapy and follow-up might be considered in the treatment plan of DM2 patients, especially those who have poor metabolic control despite implementation of numerous medical interventions. Preventive periodontal regimens for diabetic patients should be sufficiently intense and sustained as to eliminate periodontal inflammation and should be closely coordinated with the patient's overall clinical diabetic management.

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REFERENCES

1. Iacopino AM. Periodontitis and diabetes interrelationships: role of inflammation. Ann Periodontol 2001;6:125–137.
2. Rodrigues DC, Taba MJ, Novaes AB, Souza SL, Grisi MF. Effect of non-surgical periodontal therapy on glycemic control in patients with type 2 diabetes mellitus. J Periodontol 2003;74: 1361–1367.
3. Grossi SG, Skrepcinski FB, DeCaro T, *et al.* Treatment of periodontal disease in diabetics reduces glycated hemoglobin. J Periodontol 1997;68:713–719.
4. Iacopino AM, Cutler CW. Pathophysiological relationships between periodontitis and systemic disease: recent concepts involving serum lipids. J Periodontol 2000;71:1375–1384.
5. Löe H. Periodontal disease: the sixth complication of diabetes mellitus. Diabetes Care 1993;16:329–334.
6. Taylor GW, Borgnakke WS. Periodontal disease: associations with diabetes, glycemic control and complications. Oral Dis 2008;14:191–203.
7. Patiño Marín N, Loyola Rodríguez JP, Medina Solis CE, *et al.* Caries, periodontal disease and tooth loss in patients with diabetes mellitus types 1 and 2. Acta Odontol Latinoam 2008;21:127–133.
8. Wang TT, Chen TH, Wang PE, *et al.* A population-based study on the association between type 2 diabetes and periodontal disease in 12,123 middle-aged Taiwanese (KCIS No. 21). J Clin Periodontol 2009;36:372–379.
9. Nesse W, Linde A, Abbas F, *et al.* Dose-response relationship between periodontal inflamed surface area and HbA1c in type 2 diabetics. J Clin Periodontol 2009;36:295–300.
10. Grossi SG. Treatment of periodontal disease and control of diabetes: an assessment of the evidence and need for future research. Ann Periodontol 2001;6:138–145.
11. Lalla E, Lamster IB, Stern DM, Schmidt AM. Receptor for advanced glycation end products, inflammation, and accelerated periodontal disease in diabetes: mechanisms and insights into therapeutic modalities. Ann Periodontol 2001;6:113–118.
12. Taylor GW. Bidirectional interrelationships between diabetes and periodontal diseases: and epidemiologic perspective. Ann Periodontol 2001;6:99–112.
13. Katz J, Bhattacharyya I, Farkhondeh-Kish F, Perez FM, Caudle RM, Heft MW. Expression of the receptor of advanced glycation end products in gingival tissues of type 2 diabetes patients with chronic periodontal disease: a study utilizing immunohistochemistry and RT-PCR. J Clin Periodontol 2005;32:40–44.
14. Takeda M, Ojima M, Yoshioka H, *et al.* Relationship of serum advanced glycation end products with deterioration of periodontitis in type 2 diabetes patients. J Periodontol 2006;77: 15–20.
15. Lim LP, Tay FB, Sum CF, Thai AC. Relationship between markers of metabolic control and inflammation on severity of periodontal disease in patients with diabetes mellitus. J Clin Periodontol 2007;34:118–123.
16. Southerland JH, Taylor GW, Offenbacher S. Diabetes and periodontal infection: making the connection. Clin Diabetes 2005;23:171–178.
17. Dağ A, Firat ET, Arikan S, Kadiroğlu AK, Kaplan A. The effect of periodontal therapy on serum TNF-alpha and HbA1c levels in type 2 diabetic patients. Aust Dent J 2009;54:17–22.
18. Mealey BL. Periodontal disease and diabetes: a two-way street. J Am Dent Assoc 2006; 137 Suppl:26–31. Erratum J Am Dent Assoc 2008;139:252.
19. Christgau M, Palitzch KD, Schmalz G, Kreiner U, Frenzel S. Healing response to non-surgical periodontal therapy in patients with diabetes mellitus: clinical, microbiological, and immunologic results. J Clin Periodontol 1998;25:112–124.
20. Nishimura F, Murayama Y. Periodontal inflammation and insulin resistance–lessons from obesity. J Dent Res 2001;80:1690–1694.
21. Stewart JE, Wager KA, Friedlander AH, Zadeh HH. The effect of periodontal treatment on glycemic control in patients with type 2 diabetes mellitus. J Clin Periodontol 2001;28:306–310.
22. Iwamoto Y, Nishimura F, Nakagawa M, *et al.* The effect of antimicrobial periodontal treatment on circulating tumor necrosis factor-alpha and glycated hemoglobin level in patients with type 2 diabetes. J Periodontol 2001;72:774–778.
23. Faria-Almeida R, Navarro A, Bascones A. Clinical and metabolic changes after conventional treatment of type 2 diabetic patients with chronic periodontitis. J Periodontol 2006;77:591–598.
24. Navarro-Sanchez AB, Faria-Almeida R, Bascones-Martinez A. Effect of non-surgical periodontal therapy on clinical and immunological response and glycaemic control in type 2 diabetic patients with moderate periodontitis. J Clin Periodontol 2007;34:835–843.
25. Darré L, Vergnes JN, Gourdy P, Sixou M. Efficacy of periodontal treatment on glycaemic control in diabetic patients: a meta-analysis of interventional studies. Diabetes Metab 2008;34: 497–506.
26. Garcia R. Periodontal treatment could improve glycemic control in diabetic patients. Evid Based Dent 2009;10:20–21.
27. Simpson TC, Needleman I, Wild SH, Moles DR, Mills EJ. Treatment of periodontal disease for glycaemic control in people with diabetes. Cochrane Database Syst Rev 2010 May 12;5: CD004714.
28. Westfelt E, Rylander H, Blohme G, Jonasson P, Lindhe J. The effect of periodontal therapy in diabetics. Results after 5 years. J Clin Periodontol 1996;23:92–100.
29. Al-Mubarak S, Ciancio S, Aljada A, Mohanty P, Ross C, Dandona P. Comparative evaluation of adjunctive oral irrigation in diabetics. J Clin Periodontol 2002;29:295–300.
30. Madden TE, Herriges B, Boyd LD, Laughlin G, Chiodo G, Rosenstein D. Alterations in HbA1c following minimal or enhanced non-surgical, non-antibiotic treatment of gingivitis or mild periodontitis in type 2 diabetic patients: a pilot trial. J Contemp Dent Pract 2008;9:9–16.
31. Haghdoost AA, Rezazadeh-Kermani M, Sadghirad B, Baradaran HR. Prevalence of type 2 diabetes in the Islamic Republic of Iran: systematic review and meta-analysis. East Mediterr Health J 2009;15:591–599.
32. Armitage GC. Development of a classification system for periodontal diseases and conditions. Ann Periodontol 1999;4: 1–6.
33. O'Leary TJ, Drake RB, Jividen GJ, Allen MF. The incidence of recession in young males: relationship to gingival and plaque scores. SAM-TR-67-97. Tech Rep SAM-TR. 1967 Jul:1-4.
34. Löe H, Silness J. Periodontal disease in pregnancy. I. Prevalence and severity. Acta Odontol Scand 1963;21:533–51.
35. da Cruz GA, de Toledo S, Sallum EA, *et al.* Clinical and laboratory evaluations of non-surgical periodontal treatment in subjects with diabetes mellitus. J Periodontol 2008;79:1150–1157.
36. Tervonen T, Karjalainen K, Knuuttila M, Huumonen S. Alveolar bone loss in type 1 diabetic subjects. J Clin Periodontol 2000; 27:567–571.
37. Kiran M, Arpak N, Unsal E, Erdoğan MF. The effect of improved periodontal health on metabolic control in type 2 diabetes mellitus. J Clin Periodontol 2005;32:266–272.
38. Cutler CW, Machen RL, Jotwani R, Iacopino AM. Heightened gingival inflammation and attachment loss in type 2 diabetics with hyperlipidemia. J Periodontol 1999;70:1313–1321.
39. Lösche W, Karapetow F, Pohl A, Pohl C, Kocher T. Plasma lipid and blood glucose levels in patients with destructive periodontal disease. J Clin Periodontol 2000;27:537–541.

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40. Yang PS, Wang Y, Qi XM, Ren JM, Ge SH. The effect of periodontal initial therapy on circulating TNF-alpha and HbA1C in type 2 diabetes patients with periodontitis. Zhonghua Kou Qiang Yi Xue Za Zhi 2003;38:364–366.
41. Promsudthi A, Pimapansri S, Deerochanawong C, Kanchanavasita W. The effect of periodontal therapy on uncontrolled type 2 diabetes mellitus in older subjects. Oral Dis 2005;11:293–298.
42. Ryan ME. Diagnostic and therapeutic strategies for the management of the diabetic patient. Compend Contin Educ Dent 2008;29:32–38, 40-44.
43. Morrison EC, Ramfjord SP, Hill RW. Short-term effects of initial, nonsurgical periodontal treatment (hygienic phase). J Clin Periodontol 1980;7:199–211.
44. Lowenguth RA, Greenstein G. Clinical and microbiological response to nonsurgical mechanical periodontal therapy. Periodontol 2000 1995;9:14–22.
45. Badersten A, Nilvéus R, Egelberg J. Effect of nonsurgical periodontal therapy. I. Moderately advanced periodontitis. J Clin Periodontol 1981;8:57–72.

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